

WASP-52b

R-1351

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_211019 · created 2026-07-31T22:45:04+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459506.7065 +/- 0.0062 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.207 +/- 0.061
Transit depth (Rp/Rs)^2	4.3 +/- 2.53 [%]
Orbital Inclination (inc)	81.7 +/- 2.83
Airmass coefficient 1 (a1)	4.85 +/- 2.74
Airmass coefficient 2 (a2)	-1.35 +/- 1.42
Scatter in the residuals of the lightcurve fit is	58.8 %
Best Comparison Star	#2 - [572, 112]
Optimal Aperture	1.7
Optimal Annulus	3.0
Transit Duration (day)	0.039 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.207 ± 0.061 vs 0.1646 (deviation 0.7σ)
Duration fit vs published	0.039 d vs 0.0758 d (deviation 1.12σ)
Mid-time O–C	-15.4 min
Depth SNR	3.4
Residual scatter	58.8 %

Light curve

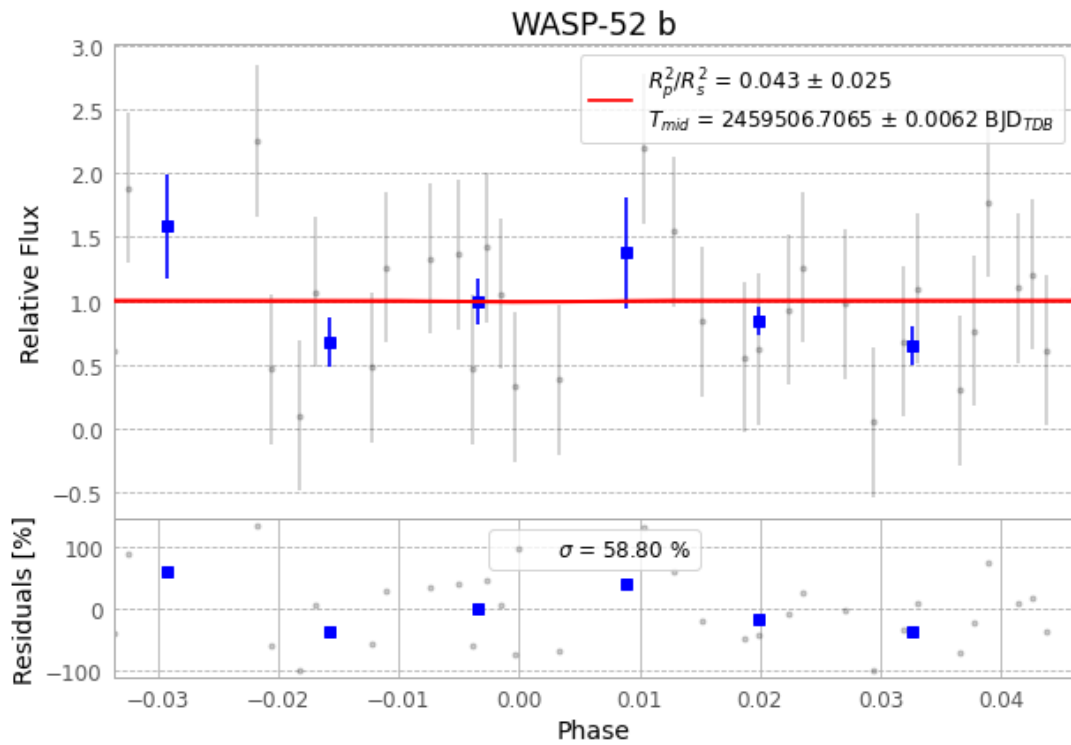


Figure 1 — FinalLightCurve_WASP-52 b_2021-10-18.png (SHA-256 a45017ff0a9e4a3d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [462, 31], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8aec0221159f2d4d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-52211019031812.FITS → WASP-52211019070612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-211019.log (398849bddb24...), FinalLightCurve_WASP-52 b_2021-10-18.png (a45017ff0a9e...), FinalLightCurve_WASP-52 b_2021-10-18.csv (869d6c062028...)

Compute resources

Wall time 420.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 22:45Z.